

Quality Management

6 — The seven statistical tools of quality

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Today

From opinions to evidence

ISO 9000 lists **evidence-based decision making** among its seven quality management principles. Today we see the toolbox that makes it possible.

Seven simple tools (actually eight) each answering **one question** about a process.

1

Why statistics

it measures variation

2

See and count

flowchart, check sheet,
Pareto chart

3

Find the causes

cause-and-effect,
stratification, scatter

4

Describe variation

samples, mean and σ ,
histogram

5

Control variation

control charts, process
capability

PART 1

Why statistics?

No two things a process makes are ever identical

Every process varies

The idea behind it

In your day

- Your commute is never exactly 25 minutes.
- Two espressos from the same bar, same barista, never taste the same.
- A “500 g” packet of pasta weighs 497 g, or 506 g.
- Your phone battery lasts a different time every day.

In a industrial process

- Two shafts turned one after the other on the same lathe differ by a few micrometres.
- Two e-scooters from the same line: one runs for years, the other dies in its first rainy winter.
- Weight, torque, temperature, solder quality: everything varies.

Unknown variation is the enemy of quality

Quality management is, in large part, **the management of variation**, and statistics is the language in which variation is described.

Two main questions: **how much** does it vary? And **why**?

Why use statistical tools?

- **Required, not optional.** ISO 9001 §9.1: decide what to monitor and measure, and how to analyse and evaluate it. §10.2: determine the causes of every nonconformity.
- **Facts, not opinions.** “The line feels slow” starts an argument. A chart of cycle times ends one — and the decision survives when its author leaves the room.
- **Causes, not symptoms.** Sorting out bad parts treats the symptom. The tools trace the effect back to the cause, so the problem does not return.
- **Priorities.** Money and people are always limited. The tools show which few problems are worth attacking first.
- **Signal versus noise.** Not every bad day is a problem. Reacting to normal variation makes a process *worse*; ignoring a real change lets it drift.
- **A common language.** A chart on the wall is read the same way by the operator, the engineer and the plant manager.

Seven tools

Kaoru Ishikawa and the Japanese quality circles

In 1960s Japan, Ishikawa (1915–1989) promoted **quality circles**: small teams of workers who meet regularly to fix problems in their own area. So the tools had to be simple — *pencil and paper*, no statistician needed.

“As much as **95 % of quality problems** can be solved with **seven fundamental tools**.”

— K. Ishikawa

His point: do not reach for advanced statistics first. Built on Shewhart (control chart, 1924), Pareto and Juran (the vital few) and Deming’s lectures in Japan (1950) — the pioneers of quality management.

The seven tools

- | | |
|---|--|
| 1 | Flowchart |
| 2 | Check sheet |
| 3 | Pareto chart |
| 4 | Cause-and-effect diagram |
| 5 | Scatter diagram |
| 6 | Histogram |
| 7 | Control chart |
| 8 | Stratification — added to the list later |

We cover all eight: the flowchart is where every investigation starts, stratification is where many are solved.

One question per tool

If you remember one slide from today, make it this one

1 • Flowchart

What actually happens, step by step?

2 • Check sheet

How often does each thing happen — when and where?

3 • Pareto chart

Which few problems cause most of the damage?

4 • Cause and effect

What could be causing this effect?

5 • Stratification

Is it the same for every machine, shift, supplier, day?

6 • Scatter diagram

Do two variables move together?

7 • Histogram

What does the variation look like?

8 • Control chart

Is the process stable over time — has something changed?

Final question

Once the process is stable, can it actually meet the specification?

Where the tools fit

PDCA and Six Sigma's DMAIC use the same toolbox

Problems are not solved in one shot. Companies follow a fixed **improvement cycle**, and repeat it until the problem is gone. Two cycles are widely used:

PDCA · Plan, Do, Check, Act

The classic cycle, from Shewhart, made popular by Deming in Japan (1950s). Plan a change, try it, check the result, then make it the standard — or start again.

DMAIC · Define, Measure, Analyse, Improve, Control

The cycle of **Six Sigma** (Motorola, 1980s). The same idea as PDCA, with the measurement steps written out.

PDCA	DMAIC	The question	Tools
	Define	What is the problem, and where in the process?	flowchart, Pareto chart
Plan	Measure	How big is it, how often, how variable?	check sheet, histogram, capability
	Analyse	Why does it happen?	cause and effect, stratification, scatter
Do	Improve	Does the fix work?	Pareto before / after, scatter, histogram
Check · Act	Control	Will it stay fixed?	control chart, capability

Whichever cycle you follow, the seven tools do most of the work in it.

The case we follow today

An e-scooter factory, and one bad week

A factory assembles **electric scooters**: hub motor, controller board and battery on one frame. Every scooter does a test ride on a roller bench at the end of the line. In one week, that test rejects **145 scooters**. Management sets a target: **reduce defects by 25 %**.

Nobody knows yet what is wrong. Opinions are plentiful:

- “It’s the new battery supplier.”
- “It’s the weekend crew.”
- “The end-of-line test has become too strict.”

We follow the investigation tool by tool. Each tool takes one step, and each step starts from the output of the one before.

The path

- 1 flowchart → where are defects found?
- 2 check sheet → which defects, which days?
- 3 Pareto → which defect first?
- 4 cause and effect → what could cause it?
- 5 stratification → is it the same every day?
- 6 scatter → is the suspect guilty?
- 7 control chart → did the fix work, and does it hold?

The **histogram** is not in the path: the case counts defects, and a histogram needs measurements. It comes in Part 4.

PART 2

See the process, count the problems

Flowchart · Check sheet · Pareto chart

Flowchart

Tool 1 · What actually happens, step by step?

- **What it is** — a diagram of the steps of a process, in order, with its decisions.
- **What it does** — shows how the work really flows, and where a defect can be caught or missed.
- **Question** — what actually happens, step by step?

Four versions of the same process

1. what management *thinks* it is
2. what the written procedure (SOP) *says*
3. what *actually happens* on the floor
4. what it *could be*

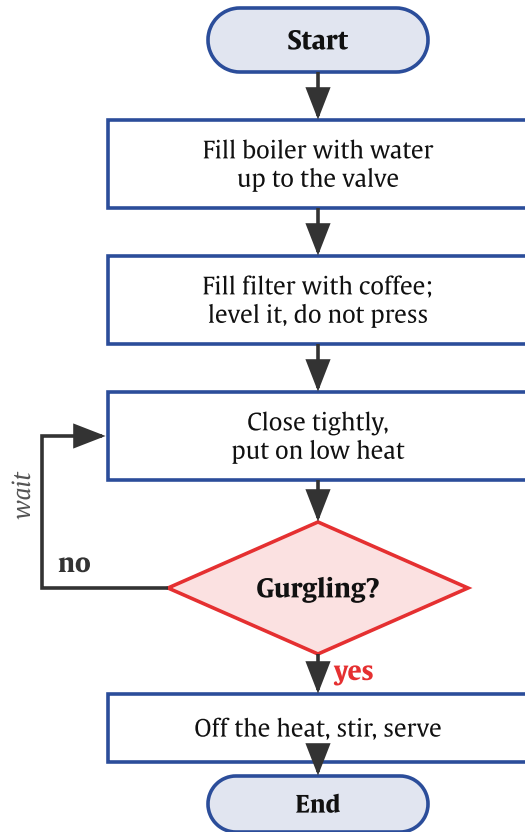
Draw number 3 first. The gap between 2 and 3 is a quality finding in itself.

“If you can’t describe what you are doing as a process, you don’t know what you’re doing.”
— W. Edwards Deming

Flowchart — two examples

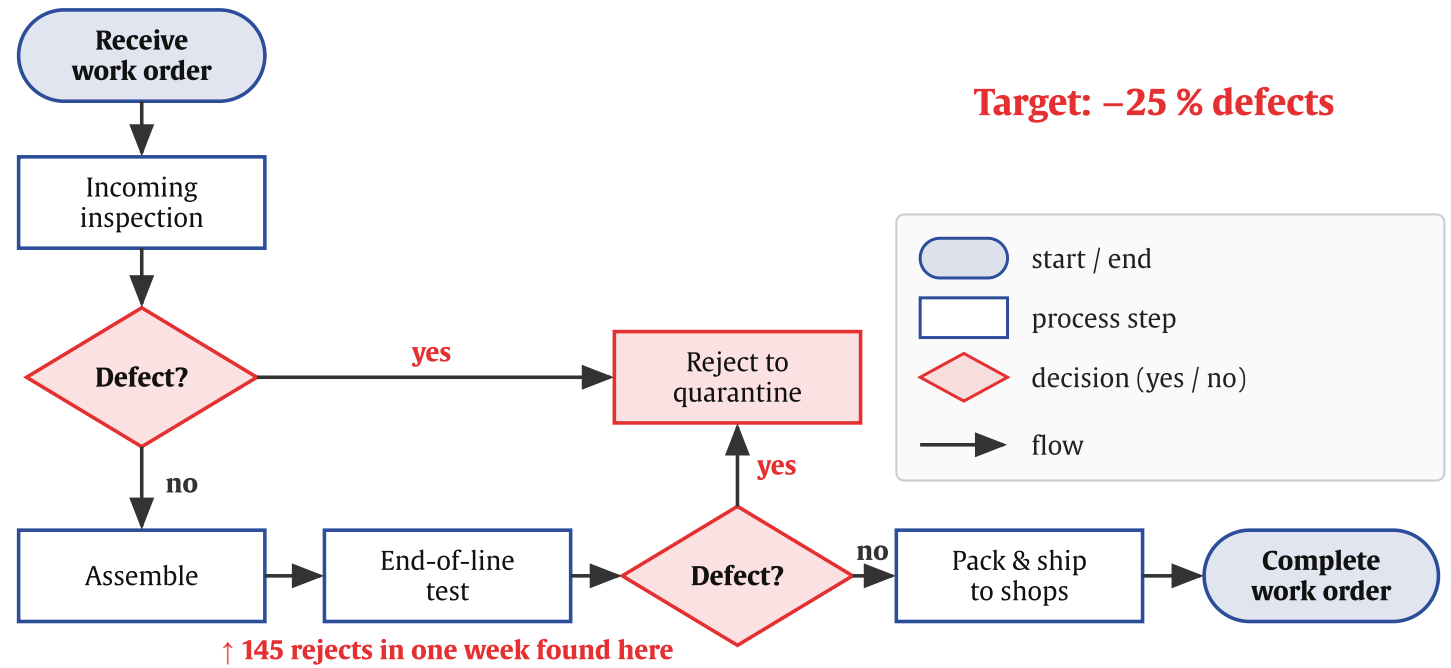
A moka pot, and the e-scooter line

EVERYDAY · COFFEE WITH A MOKA



Quality is decided at the diamond: leave it on the heat too long and the coffee burns.

QUALITY INVESTIGATION · E-SCOOTER ASSEMBLY



Two inspection points. The rejects are found at the **end-of-line test** — so that is where the data will be collected.

Check sheet

Tool 2 · How often does each thing happen — when and where?

- **What it is** — a form prepared *before* collecting, to record each event where and when it happens.
- **What it does** — turns observations into clean counts, already organised. Every other tool needs these data.
- **Question** — how often does each thing happen, when and where?

Four common types:

- **Tally sheet** — counts by category: defect type × day.
- **Location sheet** — a drawing of the part; mark where each defect is.
- **Distribution sheet** — tally measurements into ranges.
- **Checklist** — was every step done?

Metadata: data about the data

Who recorded it · **when** (date, shift, time) · **where** (line, machine). Skip it and you can never stratify later.

EVERYDAY · “WHY WAS I LATE?” — ONE SEMESTER

Reason	Mon	Tue	Wed	Thu	Fri	Total
Bus late or missed						14
Overslept						7
Looking for parking						4
Coffee queue						3
Other						2
Total	8	5	6	5	6	30

Recording *when* pays off at once: oversleeping is a **Monday** problem (4 of 7).

Check sheet — the e-scooter line

One week of end-of-line test rejects, by defect type and day

Recorded by: EOL test operator · Recorded on: week 40 · Product: city e-scooter, 350 W · Process: end-of-line (EOL) test ride · Equipment: roller bench 2

Defect	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Controller board	6	4	1	10	17	10	15	63
Hub motor	2	0	6	6	2	0	7	23
Brake noise	4	4	5	1	4	0	5	23
Connector	1	3	0	0	3	2	2	11
Hall sensor	0	0	2	3	0	2	2	9
Throttle	2	2	1	1	0	1	0	7
Battery seal	1	1	0	1	1	1	0	5
Frame scratch	0	2	1	0	1	0	0	4
Total	16	16	16	22	28	16	31	145

What the target means

145 rejects a week $\approx$ 20.7 a day.
 –25 % = **36 fewer a week**, about **5 fewer a day**.

LOOK BEFORE READING ON

Which defect dominates? Now read the *Controller board* row day by day: anything odd?

Answers

- The controller board dominates:** 63 of 145 rejects (43 %), more than the next two defects together (46).
- Its failures jump mid-week:** 11 from Sunday to Tuesday (6, 4, 1), then 52 from Wednesday to Saturday (10, 17, 10, 15). Something changed on Wednesday.

Pareto chart

Tool 3 · Which few problems cause most of the damage?

- **What it is** — a bar chart of categories sorted from largest to smallest, with a cumulative-percentage line.
- **What it does** — separates the *vital few* problems from the *useful many*, so you know what to fix first.
- **Question** — which few problems cause most of the damage?

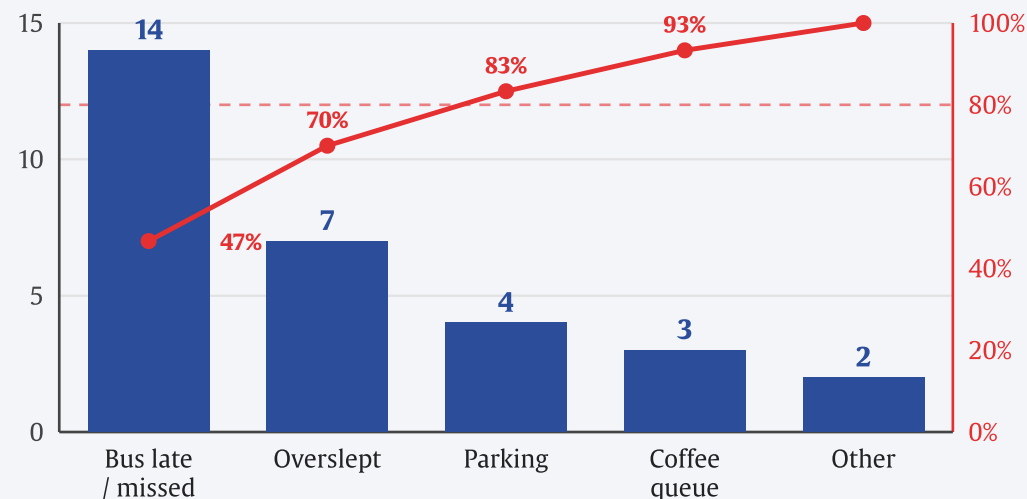
Named after V. Pareto (1896): 80 % of Italy's land owned by 20 % of the people. J. Juran brought it into quality. 80/20 is a rule of thumb, not a law.

Building one

1. Choose categories and a measure: count, cost or time.
2. Sort descending; "Other" always goes last.
3. Compute % and cumulative %.
4. Bars on the left axis, cumulative line on the right.

Excel: *Insert* → *Statistic chart* → *Pareto*, or sort and build a combo chart by hand.

EVERYDAY · THE 30 LATE ARRIVALS, SORTED

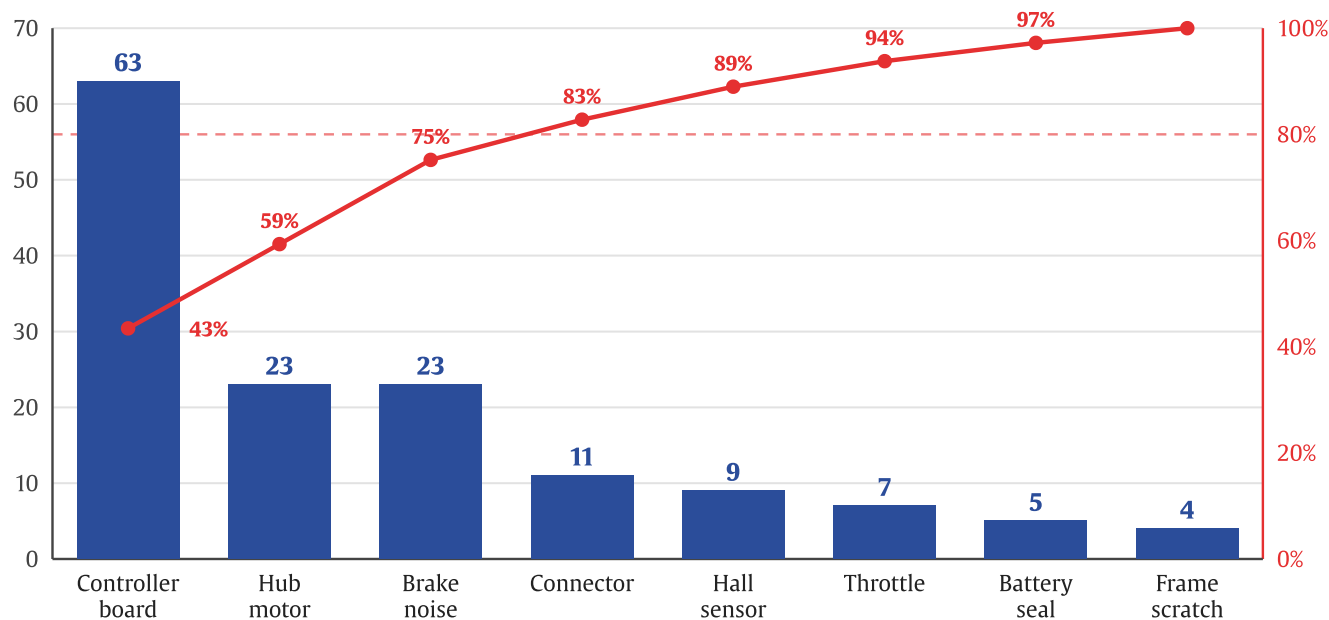


Two causes out of five explain **70 %** of the late arrivals. Leaving ten minutes earlier fixes more than any coffee strategy.

Your phone's battery screen is a Pareto chart: sorted bars, and a few apps use most of the energy.

Pareto chart — the e-scooter line

The same 145 rejects, sorted



- **Controller board alone: 43 %.**
- Top three — controller, motor, brakes — are **75 %** of all rejects, from 3 of 8 categories.
- Remove 6 in 10 controller failures and the –36 a week target is met by that alone.

Three traps

Count ≠ cost. A scratched frame is cheap to touch up, a dead controller is a new board: draw the cost Pareto too.

A huge “Other” bar: badly chosen categories.

Redraw it after the fix: before / after is the proof.

Next question: why do the controllers fail?

- 1 Every process varies. Quality management is largely the **management of variation**, and statistics is how variation is described.
- 2 The tools turn opinions into evidence, point at causes rather than symptoms, set priorities and separate signal from noise — and ISO 9001 §9.1 and §10.2 require exactly that.
- 3 **Flowchart:** draw what actually happens, not what the procedure says. The gap is itself a finding.
- 4 **Check sheet:** designed before collecting; who, when and where always recorded — or you cannot stratify later.
- 5 **Pareto chart:** sort, accumulate, attack the vital few first — and check that count and cost tell the same story.

PART 3

Find the causes

*Cause-and-effect diagram · Stratification · Scatter
diagram*

Cause-and-effect diagram

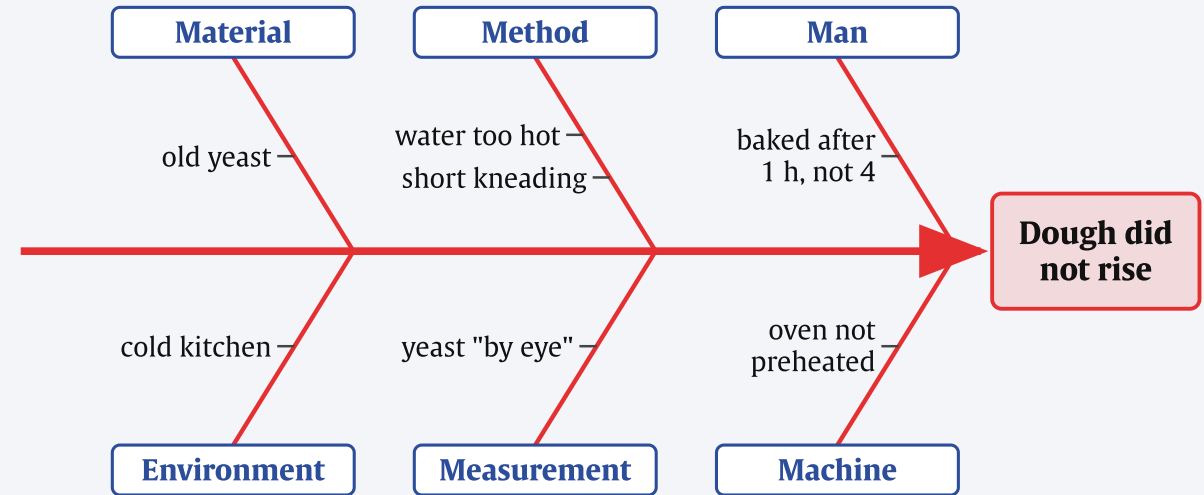
Tool 4 · What could be causing this effect? — Ishikawa, or “fishbone”

- **What it is** — a diagram with the effect at the head and the possible causes on the bones, grouped by category.
- **What it does** — collects the team’s ideas in one picture. They are hypotheses to check with data, not facts.
- **Question** — what could be causing this effect?

Building one

1. **The effect in the head** — specific and measurable: “controller failures at the end-of-line test”, not “bad quality”.
2. **The main bones** — categories of cause.
3. **Brainstorm as a team**: every idea on a bone. Ask “why?” again and again to grow smaller bones (the 5 Whys).
4. **Circle the likely causes** — then check them with data.

EVERYDAY · THE PIZZA DOUGH DID NOT RISE

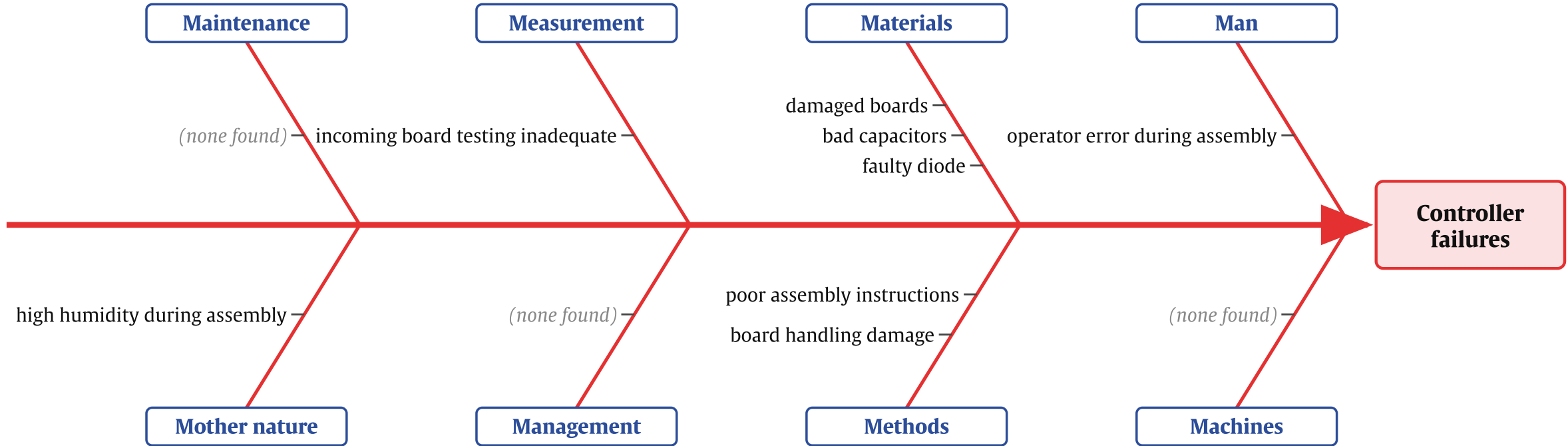


5 Whys: why no rise? yeast dead → why? water too hot → why? used the kettle to save time → why? in a hurry → why? started too late. The root cause is the *plan*, not the yeast.

6M in manufacturing: Man, Machine, Method, Material, Measurement, Mother nature (environment). **8M** adds Management and Maintenance. Services often use People, Process, Policies, Place.

Cause and effect — controller failures

The Pareto said “look at the controller”. The team asks why. (8M)



Eight candidate causes. Which one first? Some can be checked against data we already have: the check sheet recorded the *day*, and the plant logs the *humidity*.

“None found” is a result too: the team looked at that bone and found nothing. An empty bone that nobody discussed is not the same thing.

Stratification

Tool 5 · Is it the same for every machine, shift, supplier, day?

- **What it is** — splitting the data into groups by a factor that might matter.
- **What it does** — compares the groups. An average over everything can hide the answer completely.
- **Question** — is it the same for every machine, shift, supplier, day?

Typical strata: machine, operator, shift, supplier, lot, day, temperature, humidity. Only possible if the check sheet recorded who, when and where.

EVERYDAY · YOUR COMMUTE

A month of commutes averages **34 min**, from 22 to 50. Split by mode: bus 40–50 min, bike 22–28 min. The “average commute” of 34 minutes never actually happens.

INDUSTRY · A TWO-CAVITY MOULD

The parts look widely spread. Split by cavity: each cavity is tight, but one is off-centre. Fix one cavity, not the whole process.

QUALITY INVESTIGATION · CONTROLLER FAILURES BY DAY

Day	Controller failures	Humidity	Stratum
Sun	6	18 %	dry mean 3.7 / day
Mon	4	15 %	
Tue	1	12 %	
Wed	10	60 %	humid mean 13.0 / day
Thu	17	91 %	
Fri	10	72 %	
Sat	15	85 %	

All **mean 9.0 / day — describes neither group**

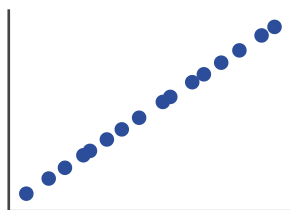
The strata are **3.5× apart**. “High humidity” from the fishbone is now the main suspect — but groups of 3 and 4 days are not proof yet.

Scatter diagram

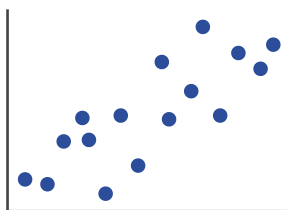
Tool 6 · Do two variables move together?

- **What it is** — a plot of pairs (x, y) from the same unit or day: x = a suspected cause, y = the effect.
- **What it does** — shows whether the two move together. The **Pearson coefficient** r , from -1 to $+1$, measures how strongly, along a straight line.
- **Question** — do two variables move together?

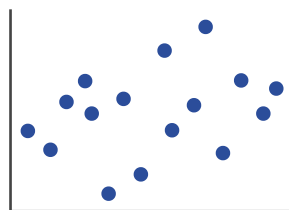
$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2]}} \quad r^2 = \text{share of the variation of } y \text{ explained by a straight line in } x$$



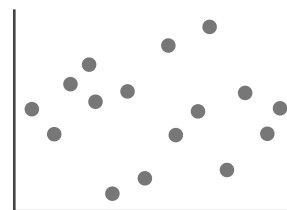
$r = +1.0$
perfect +



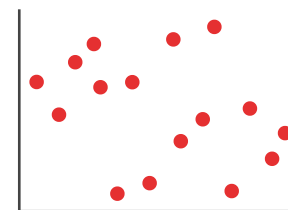
$r = +0.8$
strong +



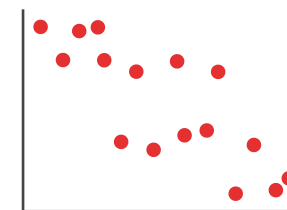
$r = +0.3$
weak +



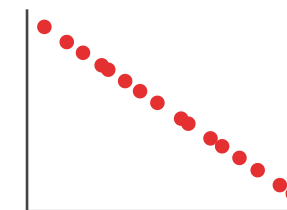
$r = 0$
none



$r = -0.3$
weak -



$r = -0.8$
strong -



$r = -1.0$
perfect -

EVERYDAY · HEATING AND THE WEATHER

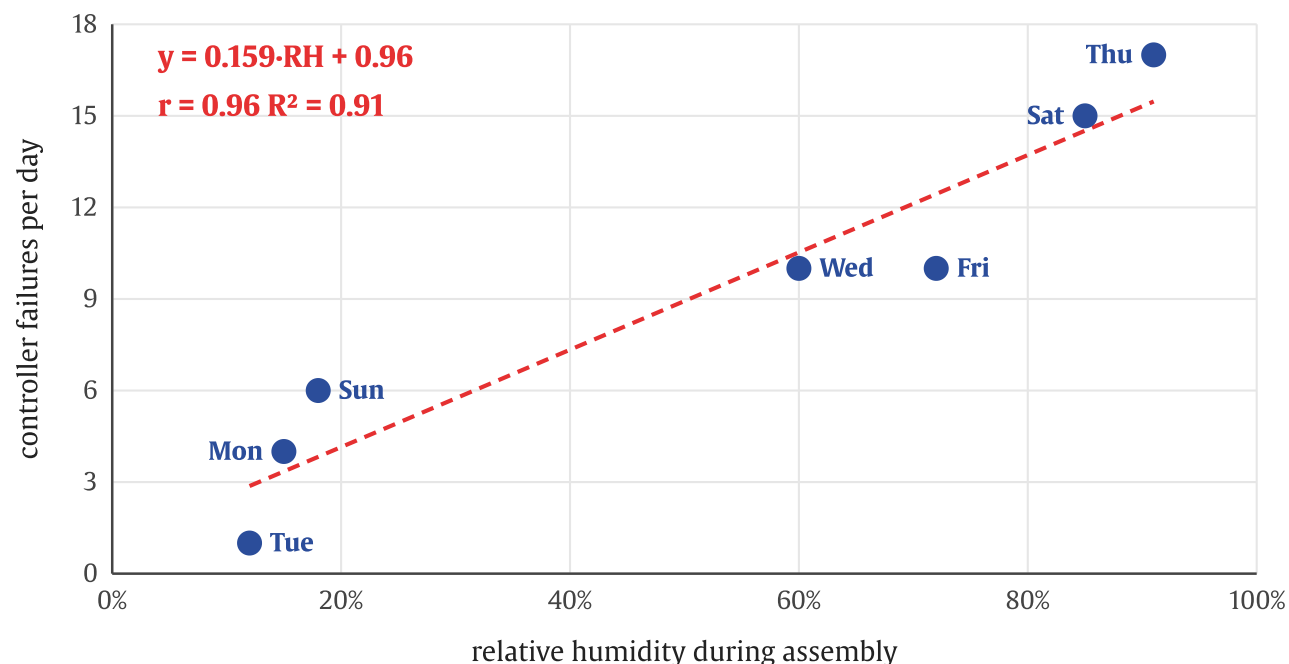
Outdoor temperature against the gas your flat burns each day: a strong **negative** r . Here the cause is real and obvious — colder days, more heating.

Correlation is not causation

Ice-cream sales and drowning accidents are positively correlated. Ice cream does not drown anyone: **summer** drives both. Always look for a hidden third variable. And $r \approx 0$ only means “no straight line” — there may still be a curve.

Scatter diagram — humidity vs controller failures

Testing the fishbone's suspect with data we already had



- $r = 0.96$, $r^2 = 0.91$: humidity accounts for about 91 % of the day-to-day variation in controller failures.
- Slope: every **+10 % humidity** $\approx$ **+1.6 failures** a day.
- A **mechanism** exists: boards and components absorb moisture before soldering — a known failure source for moisture-sensitive parts.

Still to do: change x and watch y

Control humidity in assembly, store boards dry, and see whether failures fall. That is the control chart in Part 5.

- 1 **Cause and effect:** the effect in the head, causes on the bones (6M / 8M), 5 Whys to go deeper. It produces *hypotheses*, not answers.
- 2 **Stratification:** split the data by machine, shift, supplier or condition. An overall average can describe no real group at all.
- 3 **Scatter diagram:** suspected cause on x , effect on y . r measures the strength of a straight-line relation, r^2 the share explained.
- 4 **Correlation is not causation:** look for a mechanism and a hidden third variable, and confirm by changing x .
- 5 **The tools chain:** Pareto picks the problem, the fishbone lists candidates, stratification and scatter find the real one.

PART 4

Describe variation

*Population and sample · Mean and standard deviation
· Histogram*

Population and sample

So far we counted defects. From here on we measure — and we almost never measure everything

1

Population

everything the process makes: all 10 000 scooters of a lot, size N

2

Sample

the n units you actually measure — chosen at random

3

Descriptive statistics

summarise the sample: mean, variance, standard deviation

4

Inferential statistics

from the sample, conclude something about the population

	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
Variance	$\sigma^2 = \Sigma(x - \mu)^2 / N$	$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$
Std. deviation	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$
Size	N	n

Greek letters for the population (usually unknown), Latin letters for the sample (computed).

EVERYDAY · TASTING THE SOUP

One spoonful tells you about the whole pot — **if you stir first**. Unstirred, a spoon from the top tells you about the top.

QUALITY INVESTIGATION · A LOT OF E-SCOOTERS

You measure 50 units out of 10 000. If all 50 come from the first hour of the morning shift, you have learned about the first hour, not the lot.

Mean, variance, standard deviation — by hand

Final-assembly cycle times of seven e-scooters, in seconds: 104, 126, 102, 107, 121, 105, 119

x = the final-assembly cycle time of **one scooter**, in seconds. Seven scooters, seven values. $\bar{x}$ = their mean.

x [s]	$x - \bar{x}$	$(x - \bar{x})^2$
104	-8	64
126	14	196
102	-10	100
107	-5	25
121	9	81
105	-7	49
119	7	49
$\Sigma = 784$	0	564

$$\bar{x} = 784 / 7 = 112$$

$$s^2 = 564 / (7 - 1) = 94$$

$$s = \sqrt{94} = 9.69$$

Why divide by $n - 1$?
 $\bar{x}$ is computed from the same seven numbers, so they sit closer to $\bar{x}$ than to the true μ . Dividing by n would underestimate the spread; $n - 1$ corrects it (Bessel's correction). With $n = 7$ it changes s by 8 %; with $n = 100$, by 0.5 %.

What s means
 For a bell-shaped process, about **68 %** of values lie within $\bar{x} \pm s$, **95 %** within $\pm 2s$, **99.7 %** within $\pm 3s$. Here 112 ± 9.7 holds 5 of the 7 values.

Excel: =AVERAGE(), =STDEV.S() for a sample, =STDEV.P() for a whole population.

Histogram

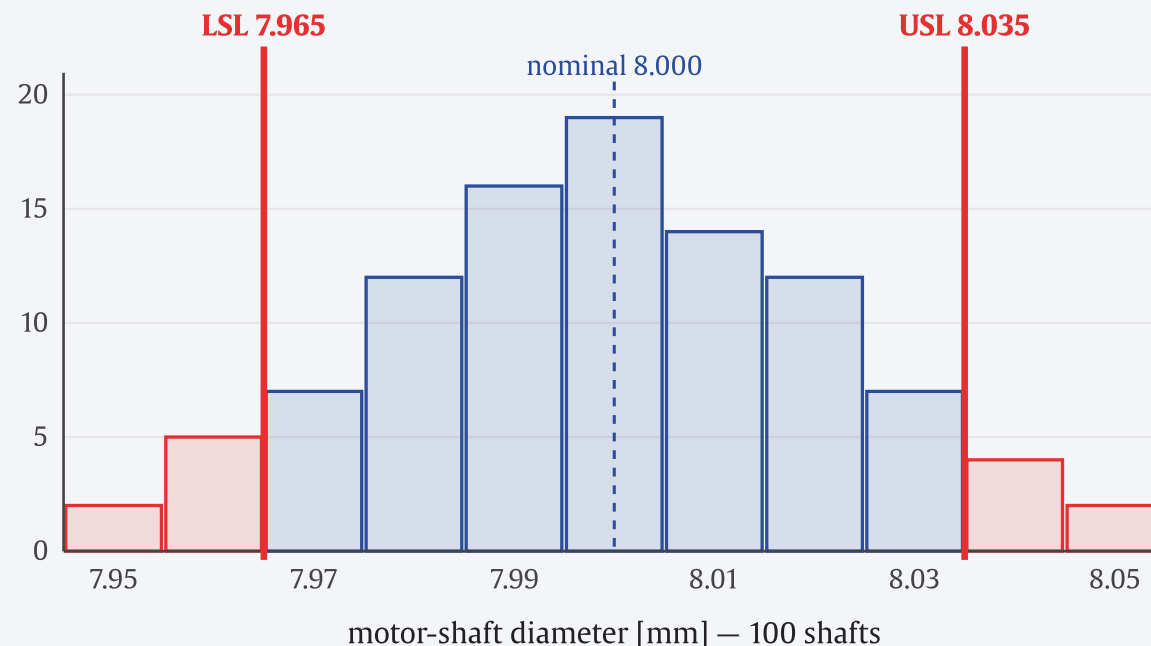
Tool 7 · What does the variation look like?

- **What it is** — a bar chart of how often *measured* values fall into equal-width classes.
- **What it does** — shows the shape of the variation (centre, spread, skew, two peaks) that a mean and an s alone would hide.
- **Question** — what does the variation look like?

EVERYDAY · 50 FOOD DELIVERIES

Most arrive in 25–30 min, with a long tail out to 50. A delivery can be very late but never very early: a **skewed** histogram, not a symmetric bell.

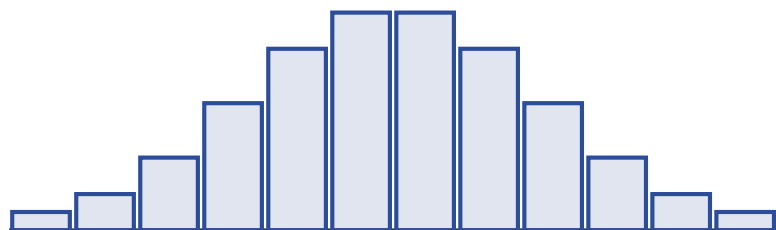
QUALITY INVESTIGATION · 100 HUB-MOTOR SHAFTS



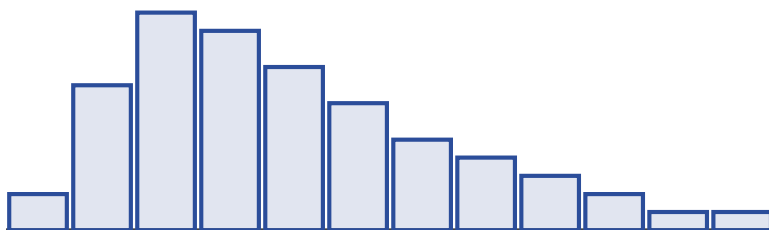
- **Bars:** how many of the 100 shafts fall in each 0.01 mm class of diameter.
- **Red lines:** the specification limits from the drawing, $\varnothing 8 \pm 0.035$ mm. **LSL** = lower specification limit (7.965), **USL** = upper specification limit (8.035). Red bars are shafts outside them.
- **Reading:** centred on nominal, but **too wide:** 13 of 100 shafts are out. Moving the centre will not help — the spread itself must shrink.

Reading a histogram

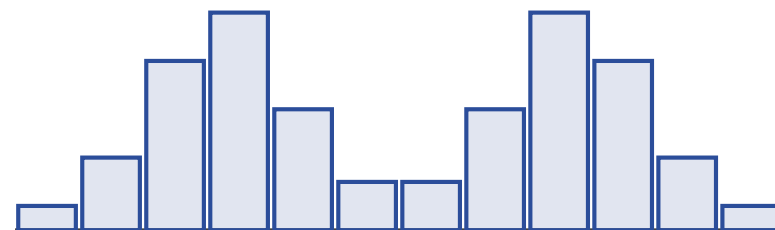
The shape tells you where to look



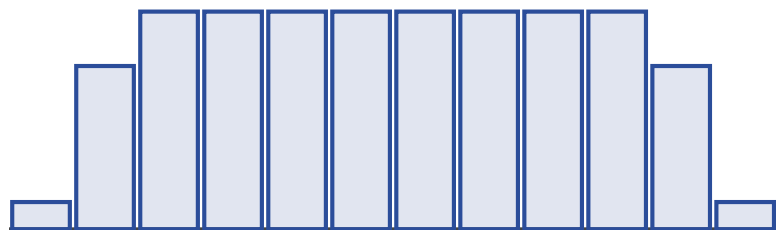
Bell (normal) — one stable process, many small independent causes. The reference case.



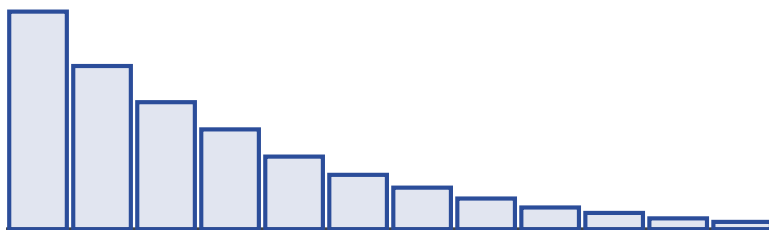
Skewed — a natural limit on one side (zero, a mechanical stop), or a tool wearing in one direction.



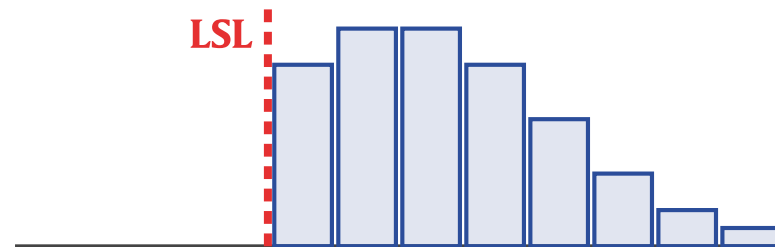
Two peaks — two populations mixed: two machines, shifts, suppliers. *Stratify*.



Plateau — several processes with different means blended, or a slow drift during the period.



Decaying — normal for times between failures or waiting times. Not every process should be a bell.



Cliff — one side cut off: out-of-spec parts were sorted out. The *process* is worse than the data you are shown.

PART 5

Control variation

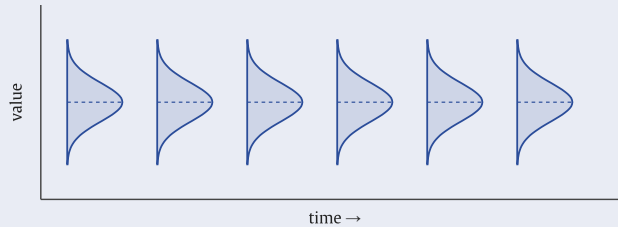
Control charts · Process capability

Two kinds of variation

Tool 8 · Control chart — Shewhart's idea, 1924

- **What it is** — a measure of the process plotted over time, with a centre line and control limits computed from the process itself.
- **What it does** — separates the two kinds of variation below: leave common causes alone, hunt down special causes.
- **Question** — is the process stable over time, or has something changed?

Common cause — a stable process

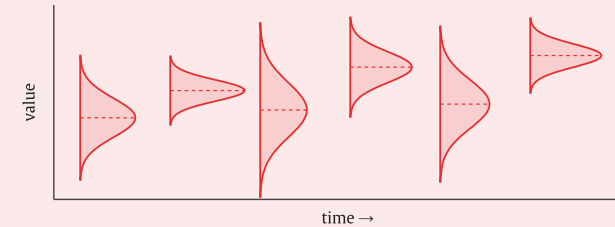


Many small causes, always present, built into the system. The output varies, but **predictably**. To reduce it you must **change the system** — a management decision.

EVERYDAY · THE BATHROOM SCALE

±0.5 kg from day to day is water and breakfast: common cause. +2 kg after the holidays, and staying there, is a special cause.

Special cause — an unstable process



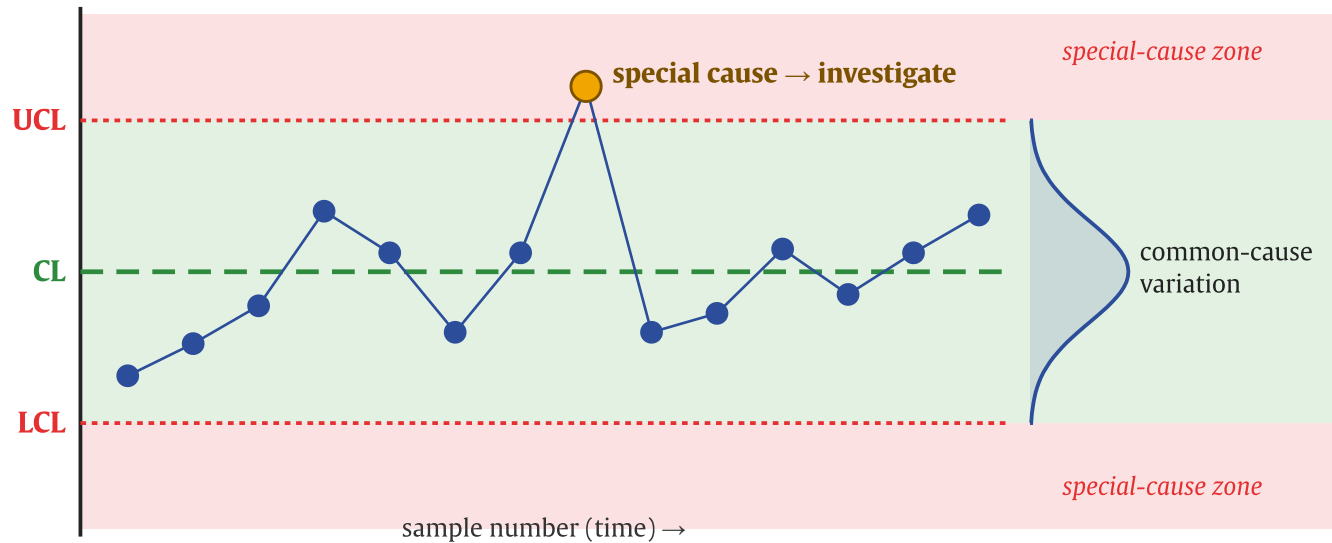
A specific, identifiable event: a worn tool, a new material lot, an untrained operator, a humid day. The output is **unpredictable**. Find the cause and **remove it**.

Two opposite mistakes

Treating common cause as special: adjusting the machine after every slightly high part. Deming's *funnel experiment* shows this **increases** variation. Or ignoring a real special cause. The control chart exists to avoid both.

Anatomy of a control chart

A centre line, and limits at $\pm 3\sigma$ of whatever is plotted



- **CL** — centre line: the process mean, from data.
- **UCL, LCL** — $CL \pm 3\sigma$. If the process is stable, 99.73 % of points fall inside; a point outside is very unlikely to be chance.

Signals of a special cause (Western Electric / Nelson rules):

- one point beyond a control limit
- 8 points in a row on one side of the CL — a shift
- 6 points in a row rising or falling — a drift
- 2 of 3 points beyond 2σ , same side

Control limits are not specification limits

Control limits are the **voice of the process**, calculated from data. Specification limits are the **voice of the customer**, written on the drawing. Never draw specification limits on an X chart: they apply to single parts, not to averages of several.

Which chart, and how to sample for it

The data type picks the chart; the subgroup sets its sensitivity

Data	Plotted	Chart
Measurements, subgroups of 2–9	mean and range	$\bar{X}$ –R
Measurements, subgroups of 10+	mean and std. dev.	$\bar{X}$ –s
Measurements, one at a time	value and moving range	I–MR
Defective units (pass / fail)	fraction or number defective	p / np
Defects (counts)	defects per unit or per period	u / c

You measure it → $\bar{X}$ –R

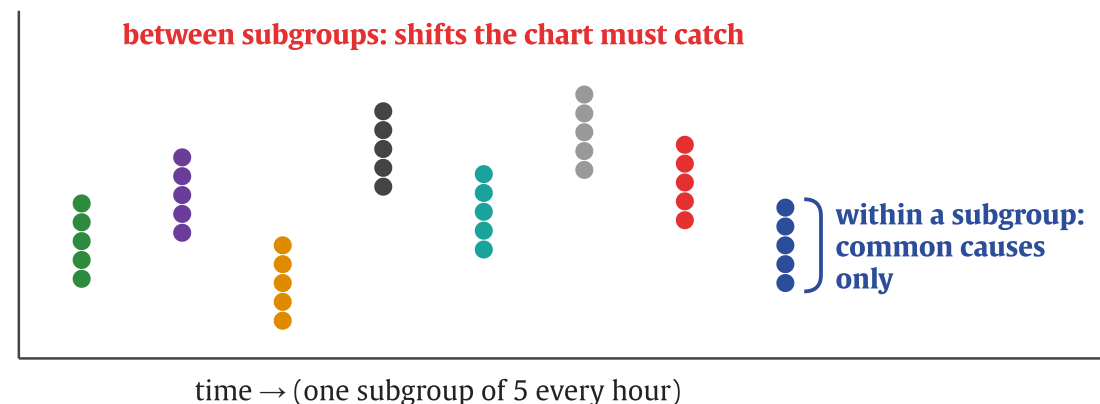
A value on a scale (amperes, mm, seconds). Every hour take a small subgroup, e.g. 3 scooters. $\bar{X}$ chart: has the *centre* moved? R chart: has the *spread* changed?

Our use: motor current.

You count it → c

Whole numbers of defects (0, 1, 2...) in a fixed amount of product or time. One point per period: the count.
Our use: controller failures per day.

Rational subgroup: a few units made under the same conditions — e.g. 3 to 5 consecutive parts, taken every hour.



The range *within* a subgroup then holds only common-cause variation. The limits are built from it, so they are tight enough to catch shifts *between* subgroups.

Wrong: one part from each of two machines in a subgroup. The machine difference inflates $\bar{R}$, the limits widen, and the chart goes blind.

$\bar{X}$ –R chart: the formulas

k subgroups of n units each

$\bar{X}$ chart — watches the process mean

$$CL = \bar{\bar{X}} = \Sigma \bar{X}_j / k \quad UCL = \bar{\bar{X}} + A_2 \bar{R} \quad LCL = \bar{\bar{X}} - A_2 \bar{R}$$

R chart — watches the spread within the process

$$CL = \bar{R} = \Sigma R_j / k \quad UCL = D_4 \bar{R} \quad LCL = D_3 \bar{R}$$

Estimate of the process σ

$$\sigma \approx \bar{R} / d_2$$

A_2, D_3, D_4, d_2 depend only on n . They turn the average range into $\pm 3\sigma$ limits, so nobody on the shop floor has to compute a standard deviation — the reason Shewhart used ranges in the age of pencils.

n	A_2	D_3	D_4	d_2
2	1.880	0	3.267	1.128
3	1.023	0	2.575	1.693
4	0.729	0	2.282	2.059
5	0.577	0	2.115	2.326
6	0.483	0	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078
15	0.223	0.347	1.653	3.472
20	0.180	0.415	1.585	3.735
25	0.153	0.459	1.541	3.931

$\bar{X}$ -R chart — worked example

Motor current at the end-of-line test, in amperes: 3 consecutive units every hour, for ten hours

Hour	1	2	3	$\bar{X}$	R
1	7	4	4	5.0	3
2	6	3	8	5.7	5
3	9	10	12	10.3	3
4	7	8	4	6.3	4
5	11	6	10	9.0	5
6	7	4	7	6.0	3
7	10	8	11	9.7	3
8	10	11	10	10.3	1
9	5	11	9	8.3	6
10	8	4	6	6.0	4
Average				7.7	3.7

① **Each hour:** mean and range. Hour 1: $(7 + 4 + 4) / 3 = 5.0$;
 $R = 7 - 4 = 3$.

② **Centre lines:** $\bar{\bar{X}} = 7.7$ $\bar{\bar{R}} = 37 / 10 = 3.7$

③ **Constants for $n = 3$:** $A_2 = 1.023$, $D_3 = 0$, $D_4 = 2.575$, $d_2 = 1.693$

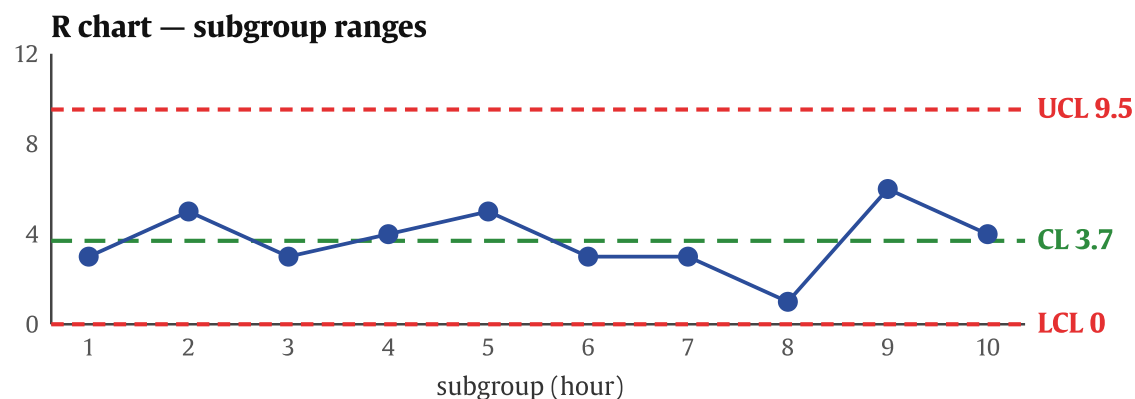
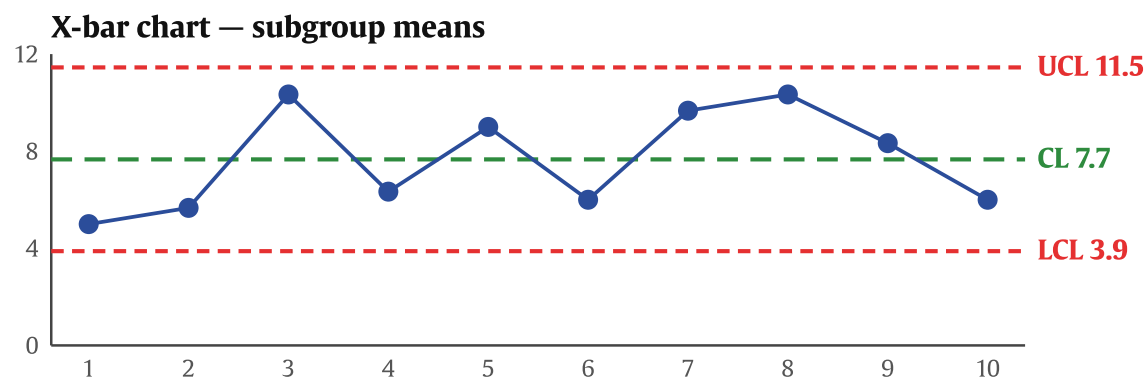
④ **$\bar{X}$ limits:** $7.7 \pm 1.023 \times 3.7 \rightarrow \text{UCL} = 11.5, \text{LCL} = 3.9$

⑤ **R limits:** $2.575 \times 3.7 \rightarrow \text{UCL} = 9.5$; $0 \times 3.7 \rightarrow \text{LCL} = 0$

⑥ **Process σ :** $3.7 / 1.693 \approx 2.2$ — we will need it for capability.

$\bar{X}$ -R chart — reading it

Same data, plotted against the limits we just computed



- **Read the R chart first.** The $\bar{X}$ limits are built from $\bar{R}$: if the spread is not stable, they mean nothing.
- All ranges below 9.5; all means between 3.9 and 11.5; no run of 8 on one side, no trend of 6.
- Verdict: **only common-cause variation** — the process is stable and predictable.

Stable does not mean good

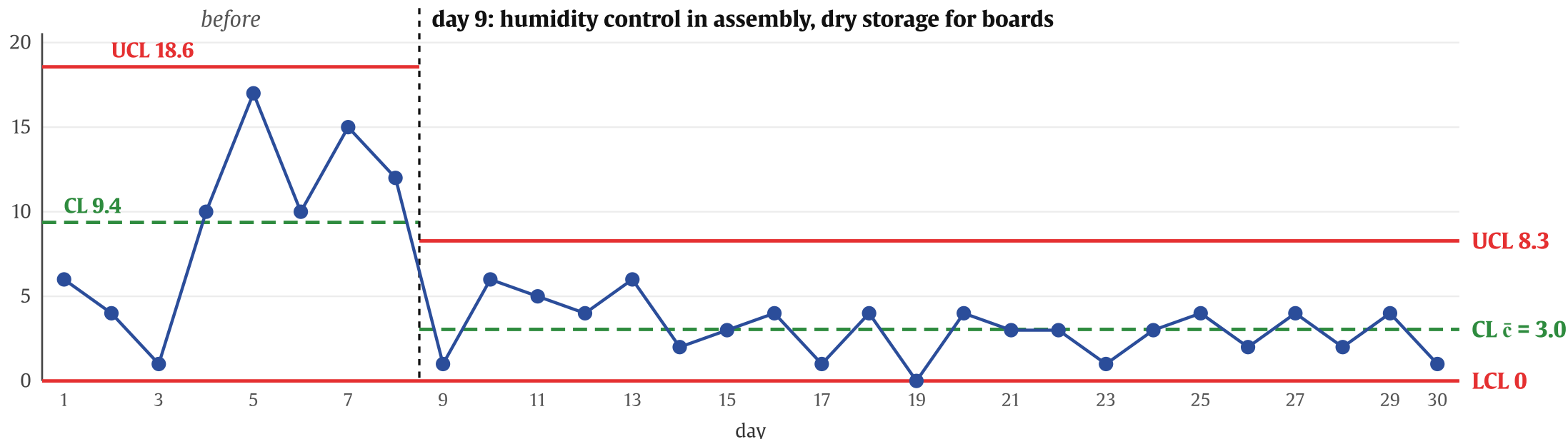
The chart says nothing about the customer. A perfectly stable process can make scrap every hour. That question is **capability**.

EVERYDAY · YOUR COMMUTE, CHARTED

One slow day is noise. Eight slow days in a row is a special cause — road works started — and a reason to change route.

Control chart — closing the e-scooter case

c chart of controller failures per day, before and after the fix



c chart (defect counts)

$$CL = \bar{c}$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL = \max(0, \bar{c} - 3\sqrt{\bar{c}})$$

For counts, the variance equals the mean. A *c* chart, not $\bar{X}$ -R: we *count* failures, we do not measure them.

Result

$\bar{c}$ from **9.4** to **3.0** failures a day: about **66** → **21** a week. One fix delivers more than the -36 target. New process, so **new limits**.

Act — hold the gain

Humidity limit written into the work instruction, dry cabinets for boards, and the chart **keeps running**.

Process capability: C_p and C_{pk}

Stable is not enough — can the process meet the specification?

- **What it is** — a number that compares the width of the specification (USL – LSL, the *voice of the customer*) with the spread of the process (6σ , the *voice of the process*).
- **Why it is used** — a stable process can still make scrap. Customers ask for it: typically $C_{pk} \geq 1.33$.
- **What it does** — tells you whether the process *can* meet the specification (C_p) and whether it *does*, where it sits now (C_{pk}).

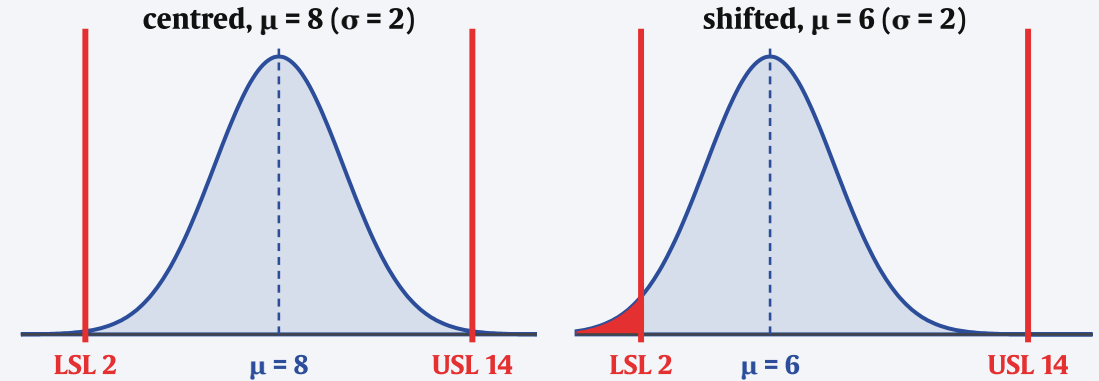
$$C_p = (USL - LSL) / 6\sigma$$

Potential: could the process fit, if it were centred?

$$C_{pk} = \min[(USL - \mu) / 3\sigma, (\mu - LSL) / 3\sigma]$$

Actual: distance to the *nearest* limit. $C_{pk} \leq C_p$, equal only when centred.

QUALITY INVESTIGATION · SAME SPREAD, DIFFERENT CENTRE



$$C_p = C_{pk} = 1.00 \cdot 0.27 \% \text{ out}$$

$$C_p = 1.00 \cdot C_{pk} = 0.67 \cdot 2.3 \% \text{ out}$$

Motor current, specification 2–14 A, $\sigma = 2$ A. $C_p = 12 / (6 \times 2) = 1.0$ in both. Shifted: $C_{pk} = \min(8/6, 4/6) = \mathbf{0.67}$. Centred: $\min(6/6, 6/6) = \mathbf{1.0}$.

Same C_p , very different scrap. Centring a process is often far cheaper than reducing its σ .

EVERYDAY · PARKING IN YOUR GARAGE

The garage width is the specification; the car plus your wobble is the process spread. C_p : how many times the car fits the garage. C_{pk} : how close you are to the nearest wall. $C_p = 2$ with $C_{pk} = 0.3$ is a big garage — and a scratched mirror.

Capability, performance, and what the numbers mean

Which σ you use changes the answer

- **C_p, C_{pk}** — σ within subgroups, $\sigma \approx \bar{R}/d_2$. Short term: what the process can do.
- **P_p, P_{pk}** — the same formulas with the overall s of all the data. Long term: what the customer actually got, drifts included.
- **$P_{pk} \approx C_{pk}$** — stable over time. **$P_{pk} \ll C_{pk}$** — the process moves between subgroups: look at the control chart.

Typical requirements

$C_{pk} \geq 1.33$ for an established process; ≥ 1.67 for new or safety-critical characteristics (common in automotive). Below 1: you are sorting, whether you admit it or not.

C_{pk}	nearest limit at	within spec	defects per million	about
0.33	1σ	68.27 %	317 311	1 in 3
0.67	2σ	95.45 %	45 500	1 in 22
1.00	3σ	99.73 %	2 700	1 in 370
1.33	4σ	99.994 %	63	1 in 16 000
1.67	5σ	99.99994 %	0.6	1 in 1.7 million
2.00	6σ	99.9999998 %	0.002	1 in 500 million

Centred, normally distributed process, both tails counted. Highlighted: our shifted example.

“Six Sigma = 3.4 defects per million”?

That figure assumes the mean drifts by 1.5σ in the long run, leaving 4.5σ to the nearest limit: it is $C_{pk} = 1.5$, not 2. Both numbers are “right” — know which convention you are reading.

- 1 **Population and sample:** μ , σ are unknown; $\bar{x}$, s are computed. The sample must be representative — stir the soup.
- 2 **Histogram:** the shape is information. Two peaks mean two processes; a cliff means somebody sorted.
- 3 **Common vs special cause:** do not tamper with a stable process; hunt down special causes.
- 4 **Control chart:** limits at $\pm 3\sigma$ come from the process, specification limits from the customer — never confuse them. Read R before $\bar{X}$.
- 5 **Capability:** C_p is what you could do, C_{pk} what you do. 1.33 is the usual minimum; only meaningful for a *stable* process.

The e-scooter case, end to end

Seven of the eight tools, one story — the histogram needs measurements, not counts

Step	Tool	What it told us
1	Flowchart	Rejects are found at the end-of-line test — collect data there.
2	Check sheet	145 rejects in a week, by defect type and by day.
3	Pareto chart	Controller failures are 43 % of rejects — start there.
4	Cause and effect	Eight candidate causes, humidity among them.
5	Stratification	Humid days: 13 failures a day. Dry days: 3.7.
6	Scatter diagram	$r = 0.96$ with humidity, and a physical mechanism to explain it.
—	The fix	Humidity control in assembly; dry storage for boards.
7	Control chart	Failures from 9.4 to 3.0 a day; new limits; chart kept running.

One question each

Each tool answered one question and handed the next tool its starting point.

Pencil and spreadsheet

None needed more than that. This is what Ishikawa meant by 95 %.

Written up as an 8D report

In industry, this story goes to the customer as an **8D report**: problem, cause, fix, proof.

Questions?

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